

Pivot-row swap at column $k=2$ (buck-like 8×8 fixture)

Before swap (col $k=2$ workspace loaded) $|x[2]| = 0.05$ too small — pivot search finds $|x[7]| = 1.4$ (highlighted yellow) **Swap rows 2 $\leftrightarrow$ 7 in x AND in stored L after 0.1 swap: column 2 ready for pivot normalisation**
 P_{row} also updated: $P_{row}[2] \leftrightarrow P_{row}[7]$

0	1							
1	-0.3	1						
2	0.1	-0.4	1					0.1
3		0.2		1				-0.1
4		-0.1			1			
5						1		0.2
6							1	
7	0.5	0.3						1.4

0	1							
1	-0.3	1						
2	0.1	-0.4	1					0.1
3		0.2		1				-0.1
4		-0.1			1			
5						1		0.2
6							1	
7	0.5	0.3						1.4

0	1							
1	-0.3	1						
2	0.5	0.3	1					1.4
3		0.2		1				-0.1
4		-0.1			1			
5						1		0.2
6							1	
7	0.1	-0.4						0.1